
IC 242: Continuum Mechanics

Course Code	IC 242
Course Name	Continuum Mechanics
L-T-P-C	2.5-0.5-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum

Course Contents

- **Introduction**
The Continuum Concept. [1 Lectures]
- **Tensor Analysis**
Tensor analysis in Cartesian coordinate, Gradient and Divergence, Daid and Daidict algebra, Isotropic Tensor [4 Lectures]
- **Stress principles**
Cauchy stress, Principle stresses and principle direction of stress, Deviatoric stresses and their directions. [10 Lectures]
- **Fluid Statics**
Pascal's law, hydrostatic pressure, pressure measurement, manometer and micro-manometer, pressure gauge. [3 Lectures]
- **Kinematics**
Lagrangean and Eulerian description, Deformation gradient, deformation tensors, strain tensors, velocity gradient, rate of deformation. [4 Lectures]
- **Conservation laws**
Conservation of mass, conservation of linear momentum, moment of momentum, conservation of energy, Integral & differential approach and application to the control volume. Clausius- Duhem equality. [8 Lectures]
- **Constitutive theories**
Governing equations of a Continuum: Constitutive equations in material description, Elastic materials, Viscous fluids, Thermodynamic considerations [8 Lectures]
- **Elasticity**
linear elasticity and hyperelasticity (compressible and incompressible materials) [2 Lectures]
- **Plasticity**
Yield criteria, linear plasticity [2 Lectures]

References

1. D. Frederick and T.S. Chang, **Continuum Mechanics** .
2. Philip G. Hodge, JR, **Continuum Mechanics** , Mc. Graw- Book Co.
3. A. C. Eringen, **Mechanics of Continuuua**, John Wiley & Sons.
4. chang, **Continuum Mechanics**, Prentice Hall, 1983.
5. Thomas, **Continuum Mechanics for Engineers**, CRC Press, 1999.
6. T. Mase, G. Mase, **Continuum Mechanics for Engineers**, CRC Press, 1999.
7. RM Bowen, **Introduction to Continuum Mechanics for Engineers**, Plenum Press, 1989.